

MRI Dataset Analysis and Preprocessing Report

Brain and Spine MRI Datasets

Abstract

Magnetic Resonance Imaging (MRI) is widely used for diagnosing brain and spine abnormalities because it provides excellent soft tissue visualization without ionizing radiation. The objective of this project is to analyze, preprocess, enhance, and prepare Brain and Spine MRI datasets for AI-based lesion Region of Interest (ROI) segmentation. The work includes dataset analysis, MRI sub-modality comparison, image property assessment, preprocessing, MRI enhancement, and deep learning-based ROI segmentation. The proposed framework improves MRI quality while preserving lesion boundaries, enabling accurate and reliable lesion detection.

1. Introduction

MRI scans acquired using different sequences such as T1, T2, FLAIR, and STIR provide complementary information for identifying abnormal tissues. Before training AI models, MRI images require preprocessing to reduce noise, improve contrast, and standardize image quality. The project focuses only on lesion ROI detection and not on disease classification. The workflow consists of four stages: dataset analysis, MRI preprocessing, MRI enhancement, and AI-based ROI segmentation.

2. Dataset Analysis

Two datasets are considered:

Brain MRI Dataset

- Modalities: T1, T2, FLAIR, T1CE
- Contains brain MRI volumes with lesion masks.
- Used for brain lesion segmentation.

Spine MRI Dataset

- Modalities: T1, T2, STIR
 - Contains spinal MRI volumes with lesion annotations.
 - Used for spine lesion segmentation.
-

3. MRI Sub-modality Comparison

Modality Main Purpose

T1	Anatomical structure visualization
T2	Detects edema and fluid
FLAIR	Highlights brain lesions by suppressing cerebrospinal fluid (CSF)
STIR	Suppresses fat signal and highlights spinal lesions

Different MRI sub-modalities provide complementary information that improves lesion localization and segmentation accuracy.

4. Dataset Challenges

The datasets contain several challenges:

- Image noise
- Low contrast
- Motion artifacts
- Different scanner resolutions
- Intensity variations
- Small lesion regions
- Class imbalance

These issues affect segmentation accuracy and require preprocessing before AI model training.

5. MRI Image Property Assessment

Before preprocessing, each MRI image is evaluated using the following image properties:

- Contrast
- Sharpness
- Edge Strength
- Noise Level

- Mean Intensity
- Standard Deviation
- Image Complexity

These parameters provide baseline information about image quality and are recalculated after preprocessing to measure improvement.

6. MRI Preprocessing (Stage 2)

The preprocessing pipeline includes:

- Image resizing
- Intensity normalization
- Noise reduction
- Artifact correction
- Contrast enhancement using CLAHE
- Data augmentation

These preprocessing techniques improve image quality while preserving lesion boundaries and reducing scanner-dependent variations.

After preprocessing, MRI quality is evaluated using the following quantitative metrics:

- Peak Signal-to-Noise Ratio (PSNR)
- Structural Similarity Index (SSIM)
- Mean Squared Error (MSE)
- Root Mean Square Error (RMSE)
- Feature Similarity Index (FSIM)
- Visual Information Fidelity (VIF)
- Blind/Reference-less Image Spatial Quality Evaluator (BRISQUE)
- Natural Image Quality Evaluator (NIQE)
- Perception-based Image Quality Evaluator (PIQE)
- Entropy
- Learned Perceptual Image Patch Similarity (LPIPS)

Higher PSNR, SSIM, FSIM, and VIF values along with lower MSE, RMSE, BRISQUE, NIQE, and PIQE values indicate improved MRI quality after preprocessing.

7. AI-Based MRI Enhancement (Stage 3)

After preprocessing, the curated Brain and Spine MRI datasets are used for AI-based image enhancement. The objective of MRI enhancement is to improve image clarity, preserve anatomical structures, and increase lesion visibility before segmentation.

A Convolutional Neural Network (CNN)-based enhancement model or an Autoencoder can be used for this task. These models learn to suppress noise, improve local contrast, and enhance fine structural details while maintaining the original anatomical information.

The enhancement model is evaluated using:

- Training Loss
- Validation Loss
- Learning Curve
- Convergence Epoch
- Cross-validation Accuracy

Model performance is also compared using computational metrics such as:

- Inference Latency
- Throughput
- GPU/CPU Utilization
- Memory Consumption
- Model Complexity

The enhanced MRI images are quantitatively evaluated using the image quality metrics described in Stage 2 to verify improvements in image quality.

8. AI-Based MRI ROI Segmentation (Stage 4)

The enhanced MRI datasets are used to train a U-Net-based deep learning model for lesion Region of Interest (ROI) segmentation. U-Net is selected because of its encoder-decoder architecture with skip connections, which enables accurate localization of lesions while preserving fine structural details.

The segmentation model is trained using annotated MRI images and evaluated using:

- Dice Similarity Coefficient (DSC)
- Intersection over Union (IoU)
- Precision
- Recall
- F1-Score
- Cross-validation Accuracy

The training process is monitored using:

- Training Loss
- Validation Loss
- Learning Curve
- Overfitting Gap
- Convergence Epoch

The final segmentation models are also compared based on:

- Inference Latency
- Throughput
- GPU/CPU Utilization
- Memory Consumption
- Model Complexity

The segmented lesion masks are exported in COCO JSON format for evaluation and submission.

9. Conclusion

The Brain and Spine MRI datasets were systematically analyzed, preprocessed, enhanced, and prepared for AI-based lesion segmentation. MRI preprocessing techniques such as normalization, denoising, resizing, artifact correction, and CLAHE significantly improve image quality. AI-based MRI enhancement further increases lesion visibility and image clarity, providing high-quality inputs for segmentation. Finally, the U-Net segmentation model accurately identifies lesion Regions of Interest (ROI), enabling reliable automated analysis of Brain and Spine MRI images. The complete workflow

establishes a robust framework for medical image enhancement and segmentation suitable for clinical research and hackathon applications.